

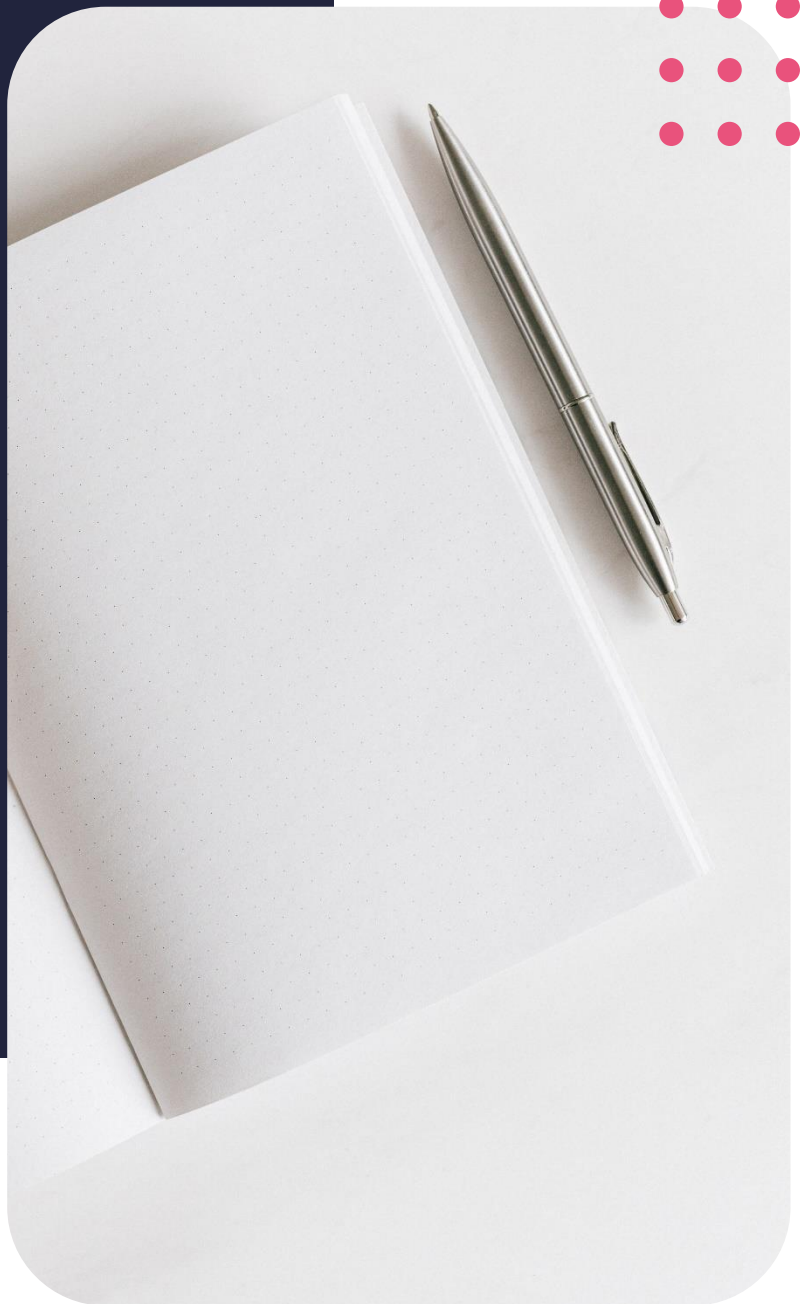
Diploma in Data Analytics

Data wrangling



Contents

3	Lesson outcomes
3	Introduction
3	Data cleaning
5	Described in R
6	Merging datasets
6	Conclusion
6	Additional resources
7	References



Lesson outcomes

By the end of this lesson, you should be able to:

- Clean data with tidyverse library
- Reproduce the descriptive results in R achieved in Module 1 with Excel
- Merge datasets

Introduction

According to director of education at RStudio, Carl Howe, being able to produce code that is reproducible so that others can build on what you have done is one of the key skills for data analysts. Howe says that the reality is many businesses run on Excel data and a good data analyst can build a dialog with end users and find ways to work with those users, and in many businesses, that may mean working with Excel data. Howe says that the problem with Excel, though, is that it guarantees that you are doing the analysis manually and that is not a recipe for creating reproducible results. Howe also states that most analytics is not doing one-offs, but doing things repeatedly, and you always want to do them in the same way and that the trick usually is to get that data out of Excel as quickly as possible and put it into a form more amenable to reproducible analysis.

Data cleaning

Data cleaning is one of the most important steps in the data analysis process. A lot of time in the data analyst journey is spent focusing on extracting useful insights from the data, but in order to extract useful and valuable insights from data, the data has to be treasure, not trash. Multiple sources state that the data cleaning step can take up to 80% of the data analyst's time. Therefore, it is important to find ways of cleaning the data as efficiently and effectively as possible.

Recap: packages in R definition

- In lesson 1 we learnt that R uses packages as Excel uses add-ons like the Data Analysis Toolpak.
- Packages are collections of R functions, compiled code and sample data.
- A package is a way to share code, documentation for the package, functions in the package and data sets as part of the package.

Package description

- The basic information of the package will be provided in the description file. Here you will be able to find out more about what the package can be used for, who the author is, what version the documentation belongs to, the date the package was created and updated, if the package has any dependencies on other packages, and so forth.
- `packageDescription("package")`
- `help(package = "package")`

Why tidy data

- Tidying data is necessary for structuring datasets to facilitate data analysis.
- Tidy data is easier to manipulate, model and visualize.
- Tidy data organizes the data values in the dataset to create a standard way of working that makes the data analysis process simpler to handle.

Tidyverse

Tidyverse is a collection of packages that transforms, organises, and visualizes data. Tidyverse is a set of packages designed to help you tidy the dataset.

All tidyverse packages can therefore be installed with a single command.

The packages have been designed to simplify the data analysis processes like tidying data and are developed by Hadley Wickham, chief data scientist at RStudio.

The packages work together, unlike current tools that require translation between packages. The idea is to make tidying data as interesting and effective as possible with the help of the collective packages.

Packages apart of the tidyverse library



(Akshat, 2019)

The elemental tidyverse packages include:

- ggplot2, for data visualisation.
- dplyr, for data manipulation.
 - dplyr functions include select, filter, groupby, and join to name a few,
- tidyr, for data tidying.
 - The goal of the tidyr is to tidy data in the consistent form where every column is a variable, every row is an observation, and every cell is a single value.
- readr, for data import.
 - Readr is used to import information from rectangular data structures like csv files.

- It is an quick and easy way to read or import rectangular data
- purrr, for functional programming.
- tibble, for tibbles, a modern re-imagining of data frames.
- stringr, for strings.
- forcats, for factors.

Data frames

- Most packages in R reads in data to a data frame.
- The function data.frame() generates data frames.
- Data frames is can be likened to a table with rows and columns.
- A data frame is of similar format to SQL tables.

Described in R

Recap titanic

In module 1 we used the Data Analysis Toolpak to explore and describe the titanic dataset. We were able to find out more about the minimum, maximum, mean, standard deviation and skewness, to name but a few, of the various variables in the dataset through the use of this add on feature. We were also able to create graphs in excel to better understand the distribution of the variables from the dataset. Let us now further explore the tidyverse library to see how it measures up against excel's data analysis process.

ggplot2

This package is over 10 years old now and is used by thousands of people. Ggplot2 is a visualizations package that is used to create plots and charts and other useful visualizations.

This package contains so many useful functions, from building box plots and density plots to time series plots. This package is also contained as a part of the tidyverse library and is installed when we run the code library (tidyverse) to access the packages of tidyverse.

Some of the basic functions of this package:

- Create a new ggplot
 - ggplot()
- Add components to a plot
 - '+' (<gg>) '%+%'
- Save a ggplot
 - ggsave()

A useful resource that lists all the interesting plots this package contains, can be found at:
<https://exts.ggplot2.tidyverse.org/gallery/>

Note: A package only needs to be installed once, but we must reload the library command with every session of `r` so that `r` will know what package we want to utilize.

How to get further help with `r`

Sometimes whilst we are learning how to use a new tool, it can be helpful to talk to other users that have gone through the same process. Often, as in life, you will find that a person that has gone through the same struggles as you have, will be able to offer you great advice on how to deal with the issues you are facing.

- RStudio community is a discussion board where you can ask questions related to R queries you might have should you get stuck in the learning process.
- Alternatively, the stack overflow community that some of you might already be familiar with is also a resource for seeking answers to questions particular to `r`. Just remember to tag your question as R on this platform.

Merging datasets

Often the data we want to extract information from lies in multiple sources and we need to integrate the various datasets. We use the term merging as an overarching term to describe combining data. Merging means that we combine different types of data into one data frame. Luckily for us, `r` is an expert in merging datasets

`dplyr`

Data sets can be joined through common columns using `r`'s `merge` function or through the `dplyr` join functions from `tidyverse`.

Remember: to combine data, there has to be an identical and unique variable in the datasets that will be combined.

We will use the package `dplyr`, for data manipulation of merging datasets. The `dplyr` package can help us solve data manipulation challenges through functions like `select`, `filter`, `groupby`, `summarise`, `arrange`, `join`, and `mutate` to name a few.

`Dplyr` uses SQL data syntax for its join functions. For instance, a right join means that everything is included on the right that was in the second data frame that we want to merge and all the rows that match from the right first data frame. So right join means to include all the rows of the right second dataset, but only rows that match from the first dataset.

Conclusion

“Education never ends, Watson, it is a series of lessons.” Sherlock Holmes

Additional resources

Grolemund, G. & Wickham, H., 2017, *R for Data Science*, available online at: <https://r4ds.had.co.nz/>

More on tidyverse packages: <https://www.tidyverse.org/packages/>

References

Still to Harvard ref

<https://campus.datacamp.com/courses/introduction-to-the-tidyverse/data-wrangling>

<https://www.tidyverse.org/packages/>

<https://datascienceplus.com/merging-datasets-with-tidyverse/>

<https://www.dummies.com/programming/r/how-to-combine-and-merge-data-sets-in-r/>

<https://medium.com/analytics-vidhya/a-beginners-guide-to-learning-r-with-the-titanic-dataset-a630bc5495a8>

<https://datascienceplus.com/getting-started-with-dplyr-in-r-using-titanic-dataset/>

<https://github.com/rstudio/cheatsheets/blob/master/data-import.pdf>

<https://www.analyticsvidhya.com/blog/2019/05/beginner-guide-tidyverse-most-powerful-collection-r-packages-data-science/>

<https://www.datacamp.com/community/tutorials/r-packages-guide#what>

<http://hadley.nz/>

<https://cran.r-project.org/web/packages/tidyr/vignettes/tidy-data.html>

<https://mran.microsoft.com/web/packages/tidyverse/vignettes/manifesto.html>

<https://tidyr.tidyverse.org/>

<https://www.rdocumentation.org/packages/base/versions/3.6.2/topics/data.frame>

<https://www.kaggle.com/varimp/a-mostly-tidyverse-tour-of-the-titanic>

<https://support.rstudio.com/hc/en-us/articles/200552336-Getting-Help-with-R>

[https://www.infoworld.com/article/3454356/how-to-merge-data-in-r-using-r-merge-dplyr-or-datatable.html#:~:text=dplyr%20uses%20SQL%20database%20syntax,left_join\(x%2C%20y\)%20](https://www.infoworld.com/article/3454356/how-to-merge-data-in-r-using-r-merge-dplyr-or-datatable.html#:~:text=dplyr%20uses%20SQL%20database%20syntax,left_join(x%2C%20y)%20)